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# The Guitarist's Companion:

*Keys & Chords*



By:

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*A music theory survival guide for the modern-day guitarist*

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*Hey there! I've created this short guide to give you access to the most important pieces of music theory that I think every guitar player should know. Think of it as your trusty companion. There for you whenever you're learning something new. Designed to help you think less and play more.*

## The Major Scale

*Western music is made up of twelve notes. If we just played these notes randomly, things wouldn't sound too musical. That's where the major scale comes in. It applies a pattern that organizes these notes and gets them sounding a lot more like music.*

*Just about every aspect of Western music theory comes from the major scale. That means it is incredibly important to have a complete understanding of how it works and how it's used. We'll start by looking at the formula used to create a major scale.*

## The Major Scale Formula

*To create a major scale from our twelve notes, we simply start on any note and apply a series of half-steps and whole-steps.*

**Half-Step (Semitone):** *A distance of one note. On the guitar, it's a distance of one fret.*

**Whole-Step (Tone):** *A distance of two notes. On the guitar, it's a distance of two frets.*

*The pattern of steps in the major scale formula is as follows:*

|                                  |
|----------------------------------|
| <b>W - W - H - W - W - W - H</b> |
|----------------------------------|

*So if we start on a note and go up a whole-step, take that note, then go up another whole-step, take that note, then a half-step, take that note, etc. We would end up with a major scale that is named after the note we started on. That means if we started on a C note and applied the major scale formula, we'd end up with a C major scale.*

*I'll just note that even though the major scale contains seven notes, the formula will give you eight notes. The eighth note is just the octave of the starting note. So if you start on a C note, you'll end on a C note.*

My favorite way to remember the major scale formula is with the phone number trick. In North America, our phone numbers look like this: **XXX-XXXX**

So if we use numbers instead of **Ws** and **Hs** to represent how far apart each note is, we get something that looks like this:

**2 2 1 - 2 2 2 1**

By applying this formula to every one of the twelve notes, we end up with twelve major scales. It's important to note that technically, you could have both a **sharp** and a **flat** version of each "in-between" scale (e.g. **C#** & **Db**), but this doesn't actually change the way the scale sounds or how it applies to the guitar fretboard.

## Major Scale Cheat Sheet

You'll notice that I've left out the **A#**, **D#**, and **G#** major scales. That's because these contain notes that have to be written as **double-sharp** (##). So in the vast majority of cases you'll be using the **flat** version (**A# = Bb**, **D# = Eb**, **G# = Ab**) instead.

| Scale Name     | I  | II | III | IV  | V  | VI | VII |
|----------------|----|----|-----|-----|----|----|-----|
| A Major Scale  | A  | B  | C#  | D   | E  | F# | G#  |
| Bb Major Scale | Bb | C  | D   | Eb  | F  | G  | A   |
| B Major Scale  | B  | C# | D#  | E   | F# | G# | A#  |
| C Major Scale  | C  | D  | E   | F   | G  | A  | B   |
| C# Major Scale | C# | D# | E#* | F#  | G# | A# | B#* |
| Db Major Scale | Db | Eb | F   | Gb  | Ab | Bb | C   |
| D Major Scale  | D  | E  | F#  | G   | A  | B  | C#  |
| Eb Major Scale | Eb | F  | G   | Ab  | Bb | C  | D   |
| E Major Scale  | E  | F# | G#  | A   | B  | C# | D#  |
| F Major Scale  | F  | G  | A   | Bb  | C  | D  | E   |
| F# Major Scale | F# | G# | A#  | B   | C# | D# | E#* |
| Gb Major Scale | Gb | Ab | Bb  | Cb* | Db | Eb | F   |
| G Major Scale  | G  | A  | B   | C   | D  | E  | F#  |
| Ab Major Scale | Ab | Bb | C   | Db  | Eb | F  | G   |

\* These are notes that break the **BC & EF rule**, which states that there is no **sharp** or **flat** between these notes. We do this so we can have one of each letter in a scale. For example, the **Gb major scale** already has a **B** note (the **Bb**), so we have to use **Cb** instead, even though **Cb** and **B** are the exact same note.

# Keys & Chords

To find all the chords that naturally exist inside of a key, we start off with a major scale. Every major key is based on its corresponding major scale. For example, the key of C major is based on the C major scale. And by taking the seven notes in the major scale and applying a new formula, we can quickly get every chord in that key.

## Major Minor Formula

Before applying this formula, we'll need to number the notes in our major scale from one through seven. Once we've done this, we apply the formula: Notes **one**, **four**, and **five** are major chords. Notes **two**, **three**, and **six** are minor chords. And the **seventh** note is a diminished chord. This will give us the seven naturally occurring chords in that key, which we call our "diatonic" chords.

## Relative Major & Minor Keys

Rather than doing a completely different process for minor keys, we rely on a concept called "relative" keys where every major key has a relative minor key. This can be found by locating the **SIXTH** note in any major scale. If you start that major scale from the sixth note, you'll have the relative minor scale.

Since keys are based on their corresponding scales, this would mean that you would now also have your relative minor key.

In the table starting on the next page, I've noted where we'd find our relative minor key for each of the major keys. So for example, the **key of A major** would contain the exact same chords as the **key of F# minor**. In the same way, the **key of C major** would contain the exact same chords as the **key of A minor**. And so on and so forth.

Just like with the major scales, I've left out the keys of **A#**, **D#**, and **G#** because of the awkward double-sharps. In these cases you should use the **flat** version instead:

**A# = Bb, D# = Eb, G# = Ab**

# Major Key Cheat Sheet

|               |           |           |            |           |         | Relative Minor Key |           |
|---------------|-----------|-----------|------------|-----------|---------|--------------------|-----------|
| A Major Scale | A         | B         | C#         | D         | E       | F#                 | G#        |
| Number        | I         | ii        | iii        | IV        | V       | vi                 | vii       |
| Chord         | A Major   | B Minor   | C# Minor   | D Major   | E Major | F# Minor           | G# Dim    |
| 7th Chord     | A Major 7 | B Minor 7 | C# Minor 7 | D Major 7 | E7      | F# Minor 7         | G# Min7b5 |

|                |            |           |           |            |         | Relative Minor Key |          |
|----------------|------------|-----------|-----------|------------|---------|--------------------|----------|
| Bb Major Scale | Bb         | C         | D         | Eb         | F       | G                  | A        |
| Number         | I          | ii        | iii       | IV         | V       | vi                 | vii      |
| Chord          | Bb Major   | C Minor   | D Minor   | Eb Major   | F Major | G Minor            | A Dim    |
| 7th Chord      | Bb Major 7 | C Minor 7 | D Minor 7 | Eb Major 7 | F7      | G Minor 7          | A Min7b5 |

|               |           |            |            |           |          | Relative Minor Key |           |
|---------------|-----------|------------|------------|-----------|----------|--------------------|-----------|
| B Major Scale | B         | C#         | D#         | E         | F#       | G#                 | A#        |
| Number        | I         | ii         | iii        | IV        | V        | vi                 | vii       |
| Chord         | B Major   | C# Minor   | D# Minor   | E Major   | F# Major | G# Minor           | A# Dim    |
| 7th Chord     | B Major 7 | C# Minor 7 | D# Minor 7 | E Major 7 | F#7      | G# Minor 7         | A# Min7b5 |

|               |           |           |           |           |         | Relative Minor Key |          |
|---------------|-----------|-----------|-----------|-----------|---------|--------------------|----------|
| C Major Scale | C         | D         | E         | F         | G       | A                  | B        |
| Number        | I         | ii        | iii       | IV        | V       | vi                 | vii      |
| Chord         | C Major   | D Minor   | E Minor   | F Major   | G Major | A Minor            | B Dim    |
| 7th Chord     | C Major 7 | D Minor 7 | E Minor 7 | F Major 7 | G7      | A Minor 7          | B Min7b5 |

|                |            |            |            |            |          | Relative Minor Key |           |
|----------------|------------|------------|------------|------------|----------|--------------------|-----------|
| C# Major Scale | C#         | D#         | E#*        | F#         | G#       | A#                 | B#*       |
| Number         | I          | ii         | iii        | IV         | V        | vi                 | vii       |
| Chord          | C# Major   | D# Minor   | E# Minor   | F# Major   | G# Major | A# Minor           | B# Dim    |
| 7th Chord      | C# Major 7 | D# Minor 7 | E# Minor 7 | F# Major 7 | G#7      | A# Minor 7         | B# Min7b5 |

|                       |            |            |           |            |           | Relative<br>Minor Key |          |
|-----------------------|------------|------------|-----------|------------|-----------|-----------------------|----------|
| <b>Db Major Scale</b> | <b>Db</b>  | <b>Eb</b>  | <b>F</b>  | <b>Gb</b>  | <b>Ab</b> | <b>Bb</b>             | <b>C</b> |
| Number                | I          | ii         | iii       | IV         | V         | vi                    | vii      |
| Chord                 | Db Major   | Eb Minor   | F Minor   | Gb Major   | Ab Major  | Bb Minor              | C Dim    |
| 7th Chord             | Db Major 7 | Eb Minor 7 | F Minor 7 | Gb Major 7 | Ab7       | Bb Minor 7            | C Min7b5 |

|                      |           |           |            |           |          | Relative<br>Minor Key |           |
|----------------------|-----------|-----------|------------|-----------|----------|-----------------------|-----------|
| <b>D Major Scale</b> | <b>D</b>  | <b>E</b>  | <b>F#</b>  | <b>G</b>  | <b>A</b> | <b>B</b>              | <b>C#</b> |
| Number               | I         | ii        | iii        | IV        | V        | vi                    | vii       |
| Chord                | D Major   | E Minor   | F# Minor   | G Major   | A Major  | B Minor               | C# Dim    |
| 7th Chord            | D Major 7 | E Minor 7 | F# Minor 7 | G Major 7 | A7       | B Minor 7             | C# Min7b5 |

|                       |            |           |           |            |           | Relative<br>Minor Key |          |
|-----------------------|------------|-----------|-----------|------------|-----------|-----------------------|----------|
| <b>Eb Major Scale</b> | <b>Eb</b>  | <b>F</b>  | <b>G</b>  | <b>Ab</b>  | <b>Bb</b> | <b>C</b>              | <b>D</b> |
| Number                | I          | ii        | iii       | IV         | V         | vi                    | vii      |
| Chord                 | Eb Major   | F Minor   | G Minor   | Ab Major   | Bb Major  | C Minor               | D Dim    |
| 7th Chord             | Eb Major 7 | F Minor 7 | G Minor 7 | Ab Major 7 | Bb7       | C Minor 7             | D Min7b5 |

|                      |           |            |            |           |          | Relative<br>Minor Key |           |
|----------------------|-----------|------------|------------|-----------|----------|-----------------------|-----------|
| <b>E Major Scale</b> | <b>E</b>  | <b>F#</b>  | <b>G#</b>  | <b>A</b>  | <b>B</b> | <b>C#</b>             | <b>D#</b> |
| Number               | I         | ii         | iii        | IV        | V        | vi                    | vii       |
| Chord                | E Major   | F# Minor   | G# Minor   | A Major   | B Major  | C Minor               | D# Dim    |
| 7th Chord            | E Major 7 | F# Minor 7 | G# Minor 7 | A Major 7 | B7       | C Minor 7             | D# Min7b5 |

|                      |           |           |           |            |          | Relative<br>Minor Key |          |
|----------------------|-----------|-----------|-----------|------------|----------|-----------------------|----------|
| <b>F Major Scale</b> | <b>F</b>  | <b>G</b>  | <b>A</b>  | <b>Bb</b>  | <b>C</b> | <b>D</b>              | <b>E</b> |
| Number               | I         | ii        | iii       | IV         | V        | vi                    | vii      |
| Chord                | F Major   | G Minor   | A Minor   | Bb Major   | C Major  | D Minor               | E Dim    |
| 7th Chord            | F Major 7 | G Minor 7 | A Minor 7 | Bb Major 7 | C7       | D Minor 7             | E Min7b5 |

|                       |            |            |            |           |           | Relative<br>Minor Key |            |
|-----------------------|------------|------------|------------|-----------|-----------|-----------------------|------------|
| <b>F# Major Scale</b> | <b>F#</b>  | <b>G#</b>  | <b>A#</b>  | <b>B</b>  | <b>C#</b> | <b>D#</b>             | <b>E#*</b> |
| Number                | I          | ii         | iii        | IV        | V         | vi                    | vii        |
| Chord                 | F# Major   | G# Minor   | A# Minor   | B Major   | C# Major  | D# Minor              | E# Dim     |
| 7th Chord             | F# Major 7 | G# Minor 7 | A# Minor 7 | B Major 7 | C#7       | D# Minor 7            | E# Min7b5  |

|                       |            |            |            |            |           | Relative<br>Minor Key |          |
|-----------------------|------------|------------|------------|------------|-----------|-----------------------|----------|
| <b>Gb Major Scale</b> | <b>Gb</b>  | <b>Ab</b>  | <b>Bb</b>  | <b>Cb*</b> | <b>Db</b> | <b>Eb</b>             | <b>F</b> |
| Number                | I          | ii         | iii        | IV         | V         | vi                    | vii      |
| Chord                 | Gb Major   | Ab Minor   | Bb Minor   | Cb Major   | Db Major  | Eb Minor              | F Dim    |
| 7th Chord             | Gb Major 7 | Ab Minor 7 | Bb Minor 7 | Cb Major 7 | Db7       | Eb Minor 7            | F Min7b5 |

|                      |           |           |           |           |          | Relative<br>Minor Key |           |
|----------------------|-----------|-----------|-----------|-----------|----------|-----------------------|-----------|
| <b>G Major Scale</b> | <b>G</b>  | <b>A</b>  | <b>B</b>  | <b>C</b>  | <b>D</b> | <b>E</b>              | <b>F#</b> |
| Number               | I         | ii        | iii       | IV        | V        | vi                    | vii       |
| Chord                | G Major   | A Minor   | B Minor   | C Major   | D Major  | E Minor               | F# Dim    |
| 7th Chord            | G Major 7 | A Minor 7 | B Minor 7 | C Major 7 | D7       | E Minor 7             | F# Min7b5 |

|                       |            |            |           |            |           | Relative<br>Minor Key |          |
|-----------------------|------------|------------|-----------|------------|-----------|-----------------------|----------|
| <b>Ab Major Scale</b> | <b>Ab</b>  | <b>Bb</b>  | <b>C</b>  | <b>Db</b>  | <b>Eb</b> | <b>F</b>              | <b>G</b> |
| Number                | I          | ii         | iii       | IV         | V         | vi                    | vii      |
| Chord                 | Ab Major   | Bb Minor   | C Minor   | Db Major   | Eb Major  | F Minor               | G Dim    |
| 7th Chord             | Ab Major 7 | Bb Minor 7 | C Minor 7 | Db Major 7 | Eb7       | F Minor 7             | G Min7b5 |

\* As was mentioned above, these are notes that break the BC & EF rule.

**B# = C, Cb = B, E# = F**

# Intervals

*Another foundational component of music theory is intervals. These are simply the names we give the distances between any two notes. The different combinations of these intervals create harmonies and the lovely things we know as chords.*

## Intervals Cheat Sheet

| Interval Name           | # of Semitones |
|-------------------------|----------------|
| Unison                  | 0              |
| Minor 2nd               | 1              |
| Major 2nd               | 2              |
| Minor 3rd               | 3              |
| Major 3rd               | 4              |
| Perfect 4th             | 5              |
| Aug 4th/Dim 5th/Tritone | 6              |
| Perfect 5th             | 7              |
| Minor 6th               | 8              |
| Major 6th               | 9              |
| Minor 7th               | 10             |
| Major 7th               | 11             |
| Octave                  | 12             |

*That's a wrap for this little guide. I hope you've found it to be helpful! I'd recommend printing it out or saving it on your phone so you always have it handy. Happy playing!*

*- Andrew*